Subject: ENVIRONMENTAL IMPACT ASSESSMENT

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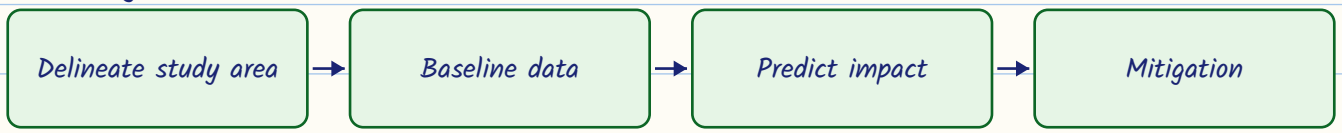
UNIT – II

Impact of Developmental Activities and Land Use

1. Introduction

Development activities such as mining, industries, dams, highways, irrigation, housing and power plants change the land, water, air and living things around them. EIA studies each component of the environment separately and then together.

- **Components studied in this unit** soil, groundwater, surface water, air and biological environment.
- For each component the general method is the same: define study area -> collect baseline data -> predict impact -> judge significance -> suggest mitigation.



Land use impact when forest or farm land is changed to industry or township, the soil structure, drainage, groundwater recharge, local climate and habitat all change.

2. Assessment of Impact on Soil and Groundwater

2.1 Introduction

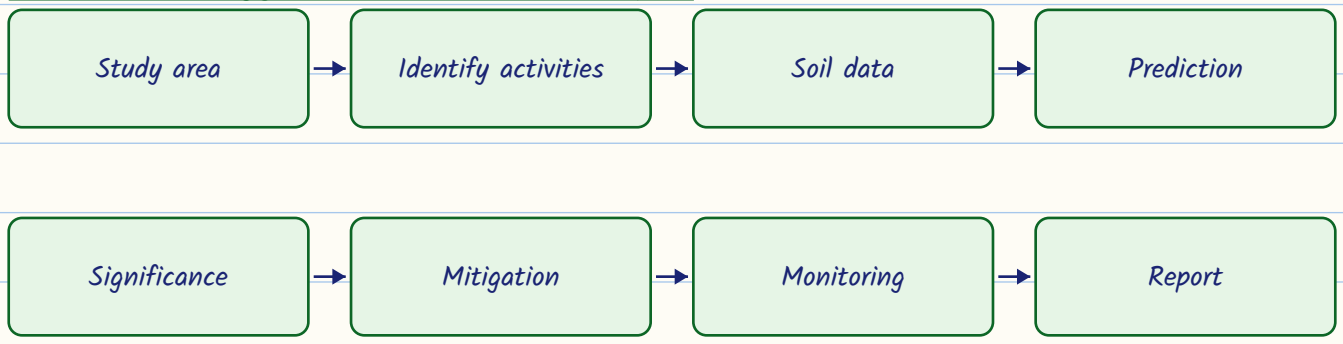
- Soil is a non-renewable resource in the human time scale; it takes hundreds of years to form a few centimetres.
- Soil supports crops, forests and filters water going down to the groundwater.

- Groundwater is the water stored in aquifers below the ground; it is used for drinking, irrigation and industry.
- Developmental activities cause soil erosion, compaction, loss of fertility, salinity, contamination and fall in groundwater table.

Common sources of impact

- **Mining and quarrying** removal of top soil, overburden dumps, acid mine drainage, lowering of water table.
- **Industries** effluent discharge, hazardous waste dumping, leakage of storage tanks.
- **Agriculture** fertilisers, pesticides, over irrigation causing waterlogging and salinity.
- **Construction** excavation, compaction, loss of recharge area due to paving.
- **Landfills** leachate entering aquifers.
- **Over pumping of borewells** depletion and seawater intrusion in coastal zones.

2.2 Methodology for Assessment (steps)



3. Delineation of the Study Area

Study area delineation means fixing the boundary of the area that may be affected by the project.

- Usually a radius of 5 to 10 km around the project site is taken (core zone is the project site and buffer zone is the 10 km radius).

- **Boundary is based on** topography, drainage pattern, groundwater flow direction, wind direction, location of villages, forests and water bodies.
- For groundwater, the catchment or the aquifer boundary is more important than a circle.
- The area is shown on the base map with the sampling stations.
- Larger area is taken for projects with larger emission or discharge, such as thermal power plants.

4. Identification of Activities

All project activities that can affect soil and groundwater are listed phase wise.

Phase	Activities	Possible effect
Pre-construction	Site survey, land acquisition, clearing	Loss of vegetation, soil exposure
Construction	Excavation, filling, transport, camps	Erosion, compaction, oil spills
Operation	Effluent, solid waste, fuel storage, water drawl	Contamination, groundwater fall
Closure / decommissioning	Dismantling, waste removal	Residual contamination

- **Activities are grouped as** land use change, material handling, waste disposal, water abstraction and accidental releases.

5. Procurement of Relevant Soil Quality Data

Baseline soil data are collected by field sampling and from secondary sources.

Secondary data

- Soil maps and reports from NBSS&LUP, State agriculture department and Survey of India toposheets.
- Land use data from remote sensing (NRSC, satellite images).
- Rainfall, geology and groundwater data from IMD, GSI and CGWB.

Primary data – soil sampling

- Samples are taken from different locations (agricultural land, forest, barren land, near the site and in the buffer zone).
- **Depth of sampling** usually 0 to 30 cm (surface); sometimes up to 90 cm in layers for profile study.
- **Quartering method** soil is collected from several points, mixed, heaped, divided into four parts and two opposite parts are rejected until about 500 g remains.
- Samples are air dried, powdered, sieved (2 mm) and sent to laboratory.

Parameters analysed

Type	Parameters
Physical	Texture, bulk density, porosity, water holding capacity, infiltration rate
Chemical	pH, electrical conductivity, organic carbon, N, P, K, cation exchange capacity, sodium adsorption ratio
Contaminants	Heavy metals (Pb, Cd, Cr, Hg), oil and grease, pesticides

Groundwater data

- **Collected from open wells and borewells** depth to water level (pre and post monsoon), pH, TDS, hardness, chloride, fluoride, nitrate, heavy metals and bacteria.
- Standards used: IS 10500 for drinking water.

6. Impact Prediction

Prediction means estimating the changes in soil and groundwater quality because of the project.

- **Soil erosion** predicted using the Universal Soil Loss Equation (USLE).

USLE $A = R \times K \times LS \times C \times P$ where A = soil loss (tonnes per hectare per year), R = rainfall erosivity, K = soil erodibility, LS = slope length and steepness factor, C = crop or cover factor, P = conservation practice factor.

- **Contamination of soil** mass balance method (input of pollutants - output = accumulation) and dispersion of air pollutants that settle on soil.
- **Groundwater level** Darcy's law for flow, $Q = K \times i \times A$, and groundwater flow or numerical models (MODFLOW).
- **Groundwater quality** contaminant transport models predict movement of leachate or effluent plume.
- **Analogy method** comparing with the results observed at a similar existing project.
- Expert opinion and checklists are used where data is not enough.
- Prediction is done for different stages and for worst case conditions.

7. Assessment of Impact Significance

After prediction, the predicted change is compared with a standard or a threshold to say whether the impact is important.

- **Compare predicted values with** IS 10500, CPCB standards, FAO soil guidelines and background (baseline) levels.
- **Consider** magnitude, extent, duration, reversibility, sensitivity of receptor (farm land, drinking water source) and number of people affected.
- **Rating given as** negligible, minor, moderate, major or severe.
- **Significant impact examples** pollutant level exceeding permissible limit, loss of prime agricultural land, fall of water table beyond natural fluctuation, contamination of a drinking water source.
- Cumulative impacts of nearby industries are also considered.

8. Identification and Incorporation of Mitigation Measures

Mitigation follows the hierarchy: Avoid -> Minimise -> Restore -> Compensate.

For soil

- Preserve and stockpile the top soil separately and reuse it for green belt and reclamation.
- Terracing, contour bunding, check dams and grassing on slopes to control erosion.
- Proper design of waste dumps with retaining walls and garland drains.
- Lined and covered storage for hazardous waste; engineered landfills with leachate collection.
- **Soil remediation methods** bioremediation, phytoremediation, soil washing and excavation.
- Afforestation and green belt with local species.

For groundwater

- Rain water harvesting and artificial recharge pits to compensate for abstraction.
- Impermeable lining of ponds, tanks and drains.
- Treatment of effluent to standards before use or disposal; zero liquid discharge where possible.
- Controlled pumping and use of recycled water.
- Monitoring wells around the site for regular quality checks.
- These measures are written into the Environmental Management Plan with cost and responsibility.

9. EIA in Surface Water Environment

9.1 Introduction

Surface water includes rivers, streams, lakes, ponds, reservoirs, estuaries and the sea. Projects affect it by discharging wastewater, changing flow, increasing siltation and altering drainage.

9.2 Sources of impact

- Industrial effluents and sewage discharge.
- Thermal discharge from power plants (thermal pollution).
- Runoff from construction sites, mines and agricultural fields (nutrients, pesticides, sediments).
- Dams and barrages changing flow, sediment movement and fish migration.
- Dredging, reclamation and port activities.
- Solid waste and oil spills.

9.3 Methodology for assessment

- **Step 1** Describe the project and list the water related activities (water drawl, effluent, runoff).
- **Step 2** Delineate the study area - the catchment, the stretch of river and downstream users.
- **Step 3** Baseline data collection - flow, depth, velocity, rainfall, and water quality.
- **Step 4** Identify water uses (drinking, irrigation, fishing, bathing, industry) and the water quality criteria for each use.
- **Step 5** Predict impacts by modelling.
- **Step 6** Evaluate significance against standards.
- **Step 7** Suggest mitigation and monitoring.

Baseline parameters

Group	Parameters
Physical	Temperature, colour, turbidity, TSS, TDS

Chemical	pH, DO, BOD, COD, nitrates, phosphates, chlorides, oil and grease, heavy metals
Biological	Coliform count, plankton, benthic organisms, fish

Prediction models

- **Dissolved oxygen sag model (Streeter-Phelps)** predicts the fall of DO downstream of a waste discharge and the critical point where DO is minimum.

Streeter-Phelps $dD/dt = K1 \times L - K2 \times D$ where D is the oxygen deficit, L is the BOD remaining, K1 is the deoxygenation rate and K2 is the reaeration rate.

- **Dilution and mixing model** $C = (Qr \times Cr + Qw \times Cw) / (Qr + Qw)$ gives the concentration after complete mixing of the river and the waste.
- Lake and reservoir models for eutrophication; hydrodynamic models for estuaries.

Mitigation

- Treat effluent (primary, secondary and tertiary) before discharge; reuse and recycle.
- Cooling towers or ponds to reduce thermal pollution.
- Silt traps and sedimentation basins at construction sites.
- Maintain minimum environmental flow downstream of dams.
- Buffer strip of vegetation along river banks.
- Regular monitoring of water quality and emergency plan for spills.

10. EIA in the Air Environment

10.1 Air pollution sources

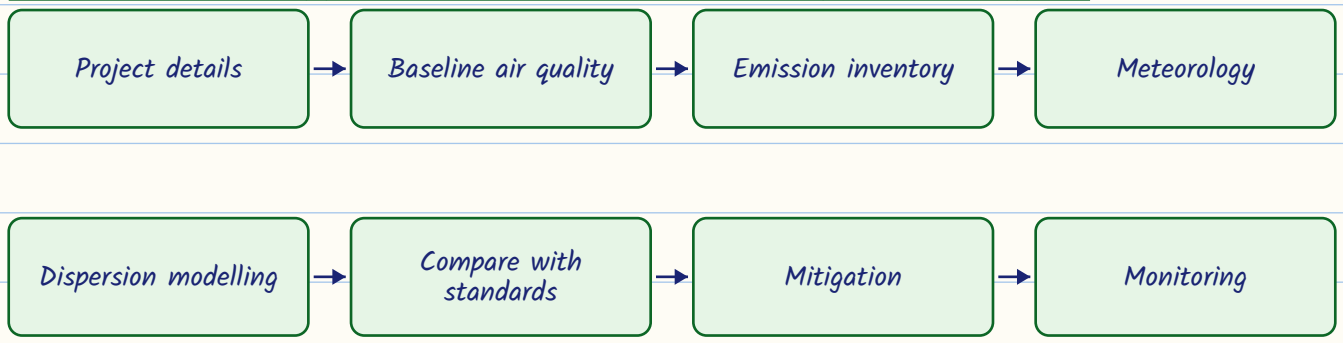
- **Point sources** chimneys and stacks of industries and power plants.
- **Line sources** roads, highways and railway lines (vehicle exhaust).
- **Area sources** open burning, mining areas, construction sites, landfills, towns.

- **Natural sources** dust storms, volcanoes, forest fires.
- **Primary pollutants** SO2, NOx, CO, SPM, PM10, PM2.5, hydrocarbons, lead.
- **Secondary pollutants** ozone, smog, acid rain formed by reactions in the atmosphere.

Effects

- **On health** respiratory diseases, heart diseases, irritation to eyes.
- **On plants** leaf damage, reduced yield. On materials :: corrosion of buildings and monuments (e.g. Taj Mahal).
- **On climate** global warming, ozone layer damage.

10.2 Generalised approach for assessment of air pollution impact



- **Step 1** Collect project details - fuel used, production, stack height, emission rate.
- **Step 2** Baseline air quality monitoring at a number of stations (upwind, downwind and in villages) for SPM, PM10, PM2.5, SO2, NOx, CO as per CPCB; monitoring for 24 hours twice a week for a season.
- **Step 3** Meteorological data - wind speed and direction, temperature, humidity, rainfall, stability class (from IMD or site station); wind rose is drawn.
- **Step 4** Emission inventory - list all sources and their emission rates.
- **Step 5** Dispersion modelling to predict ground level concentration (GLC).
- **Step 6** Add predicted concentration to baseline and compare with National Ambient Air Quality Standards (NAAQS).

- **Step 7** Plan control measures and monitoring.

Dispersion models

- **Gaussian plume model** the basic model, assumes pollutant spreads like a normal (bell shaped) curve horizontally and vertically.

Gaussian plume C(x,y,z) = [Q / (2 x pi x u x sigma-y x sigma-z)] x exp(-y squared / 2 sigma-y squared) x [exp(-(z-H) squared / 2 sigma-z squared) + exp(-(z+H) squared / 2 sigma-z squared)]

- Here Q = emission rate (g/s), u = wind speed (m/s), H = effective stack height, sigma-y and sigma-z = dispersion coefficients (depend on stability class and distance x).
- **Software commonly used** ISCST3, AERMOD, CALPUFF and the line source model CALINE for roads.
- Effective stack height H = physical stack height + plume rise.

Mitigation for air

- **Control equipment** ESP, bag filter, cyclone, scrubber for particles; flue gas desulphurisation for SO2.
- Tall stacks for dispersion; use of clean fuels (gas instead of coal).
- Dust suppression by water sprinkling and covered conveyors.
- Green belt, good housekeeping, vehicle maintenance.
- Continuous emission monitoring system (CEMS).

II. EIA in the Biological Environment (Overview)

- Biological environment means flora, fauna and ecosystems of the study area.
- **Baseline** species list, density, diversity indices (Shannon-Wiener), endangered species, forests, protected areas, migration routes.
- **Methods** quadrat and transect survey, line transect for animals, secondary data from Forest department and Zoological and Botanical Surveys.

- **Impacts** loss of habitat, fragmentation, mortality, pollution effects and invasive species.
- **Mitigation** avoid sensitive areas, compensatory afforestation, wildlife corridors, translocation and conservation plan.
- More detail is given in Unit III.

12. Quick Revision Table

Component	Key parameters	Prediction tool	Typical mitigation
Soil	pH, EC, organic carbon, metals	USLE, mass balance	Top soil conservation, terracing
Groundwater	Water level, TDS, hardness	Darcy law, MODFLOW	Recharge, lining
Surface water	DO, BOD, COD, turbidity	Streeter-Phelps, mixing model	ETP, silt traps
Air	SPM, SO2, NOx, CO	Gaussian plume, AERMOD	ESP, stack, green belt

Important questions (1) Explain the methodology for assessment of impact on soil and groundwater. (2) Describe delineation of study area. (3) Explain the steps in assessing impacts on surface water. (4) Explain the generalised approach for air pollution impact assessment. (5) Write about Gaussian plume model and mitigation of air pollution.

13. Worked Examples

Example 1 - Mixing of waste with river water

A river has flow Qr = 10 m3/s with BOD Cr = 2 mg/L. A factory discharges Qw = 0.5 m3/s of waste with BOD Cw = 200 mg/L. Find the BOD after complete mixing.

- C = (Qr x Cr + Qw x Cw) / (Qr + Qw)
- C = (10 x 2 + 0.5 x 200) / (10 + 0.5)

- C = (20 + 100) / 10.5 = 11.43 mg/L

Result BOD after mixing is about 11.4 mg/L, which is much higher than the river's original 2 mg/L. So treatment of the waste is needed.

Example 2 – Soil loss by USLE

Given R = 400, K = 0.3, LS = 1.2, C = 0.5, P = 0.8.

- A = R x K x LS x C x P
- A = 400 x 0.3 x 1.2 x 0.5 x 0.8 = 57.6 tonnes per hectare per year
- Comparing with the tolerable soil loss (about 10 to 12 t/ha/year in India), the loss is high, so conservation practices such as contour bunding and cover crops are needed.

14. Key Points to Remember

- Study area is normally a 10 km radius, but groundwater uses the aquifer or catchment boundary.
- Soil sampling is done at 0 to 30 cm depth and by the quartering method.
- **Prediction tools** USLE for erosion, Darcy and MODFLOW for groundwater, Streeter-Phelps for river DO, Gaussian plume for air.
- Significance is judged against standards (IS 10500, CPCB, NAAQS) and baseline.
- **Mitigation hierarchy** avoid, minimise, restore, compensate.
- Baseline air monitoring is done for one season, twice a week, for 24 hours.